

SLAM - Ex 7

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July 8, 2026

This document contains the answers to VAN Exercise 7. The code for this exercise is in [this link](#). The final version used for this submission corresponds to commit `ca9b8ff`.

For each current keyframe c_n , we consider earlier keyframes c_i , $i < n$, as potential loop-closure candidates.

We initially observed numerical instability caused by extremely small BA-derived covariances, which made the pose-graph constraints overly confident and reduced the effect of loop-closure edges. To handle this, we replaced the fixed 10-frame keyframe interval with a motion-aware selection rule: from a bounded range of frames ahead (11 – 18), we choose a keyframe with sufficient relative translation or rotation. This yields more meaningful motion between keyframes while keeping BA windows bounded. We also add a small covariance regularization term before using the covariances in the Mahalanobis test and pose graph.

1 Detect Loop Closure Candidates

1.1 Relative Covariance

For each candidate pair (c_i, c_n) , we approximate the relative covariance by finding a minimum-uncertainty path between the two keyframes in the current pose graph. Following the approximation described below, the covariance along a path is obtained by summing the covariances of its relative-pose measurements:

$$\Sigma_{n|i} \approx \sum_{e \in P_{i \rightarrow n}} \Sigma_e.$$

Each directed graph edge represents an existing relative-pose constraint, either from a local Bundle Adjustment window or from an already accepted loop closure. Thus, the edge weight $w(u, v)$ is the full 6×6 covariance matrix of the relative measurement from keyframe u to keyframe v .

To compare accumulated covariance matrices, we use

$$\mathcal{U}(\Sigma) = \sqrt{\det(\Sigma)}.$$

For a Gaussian distribution, this quantity is proportional to the volume of a constant-probability uncertainty ellipsoid; hence, a smaller value corresponds to a more certain

relative-pose estimate. Since the square-root function is monotone, comparing $\sqrt{\det(\Sigma)}$ is equivalent to comparing $\det(\Sigma)$.

We implemented the covariance-valued variant of Dijkstra’s. For every keyframe v , the algorithm maintains the covariance $\Sigma[v]$ of the currently best path from c_i to v . When relaxing a directed edge (u, v) , it forms

$$\Sigma_{\text{candidate}} = \Sigma[u] + w(u, v),$$

and replaces the current path to v when $\det(\Sigma_{\text{candidate}}) < \det(\Sigma[v])$. After recovering the selected path, its accumulated covariance is used as the approximation $\Sigma_{n|i}$.

1.2 Detect Possible Candidates

For each current keyframe c_n , we consider sufficiently earlier keyframes c_i . At this stage no loop-closure measurement between c_i and c_n is available yet, so we use the current pose-graph estimate only as an approximation. We compute the estimated relative pose

$$\Delta c_{ni} = \text{t2v}(C_i^{-1}C_n).$$

A candidate pair is considered plausible when its squared Mahalanobis distance,

$$d_{ni}^2 = \Delta c_{ni}^\top \Sigma_{n|i}^{-1} \Delta c_{ni},$$

is relatively small, where $\Sigma_{n|i}$ is the relative covariance estimated from the pose graph. To avoid selecting temporally adjacent frames, we exclude the five most recent keyframes.

We empirically calibrated the Mahalanobis threshold. Using temporally nonlocal pairs within 5 m and 20° as a proxy for possible candidates, the smallest threshold retaining at least 90% of these pairs was $d_{\max}^2 = 32023.8$. This threshold selected 67 candidate pairs over 25 target keyframes.

2 Consensus Matching

For each Mahalanobis-selected candidate, we performed four-view matching using the stereo correspondences in both frames and temporal matches between the two left images. We then ran PnP/RANSAC and accepted the candidate as a verified loop closure if it produced at least 20 inliers under the 2-pixel reprojection criterion used in Exercise 3.

Using this criterion, 67 Mahalanobis-selected candidates were tested, and 52 were verified as successful loop closures. Since many of these verified matches correspond to nearby frames along the same loop-overlap region, we further kept only robust and temporally spread representatives for the BA refinement stage: we required at least 30 RANSAC inliers and selected 10 representative loop closures spread along the matched overlap.

3 Relative Pose Estimation

A reasonable initial solution for the Bundle is fixing the source camera pose to identity and initializing the target camera pose using the relative transform estimated by PnP/RANSAC from the verified inlier matches. The 3D landmarks were initialized from the source stereo triangulation, so the small BA only had to refine an already geometrically valid two-frame reconstruction.

To extract the appropriate covariance we computed GTSAM marginals after optimizing the two-frame stereo BA graph, and extracted the marginal covariance of the optimized target pose.

4 Update the Pose Graph

For each refined loop closure, we added a new `BetweenFactorPose3` between the two corresponding keyframes. The measured relative pose was the one obtained from the two-frame BA in Section 7.3, and the factor noise model was constructed from the relative-pose covariance.

5 Performance Analysis

Out of 67 Mahalanobis-selected candidate pairs, 52 were detected as successful loop closures by the four-view PnP/RANSAC consensus test, using a threshold of at least 20 inliers. Since many of these loop closures were redundant matches from the same overlap regions, we used only 10, temporally spread representatives for BA refinement and pose-graph optimization.

We show one successful consensus match in Figure 1. The match is between frames 0 and 1429, with 33 inliers and 7 outliers out of 40 four-view correspondences. The figure was generated by `plot_consensus_match` in `src/slam/visualization/ex7_plots.py`, called from `q_5` in `ex7.py`.

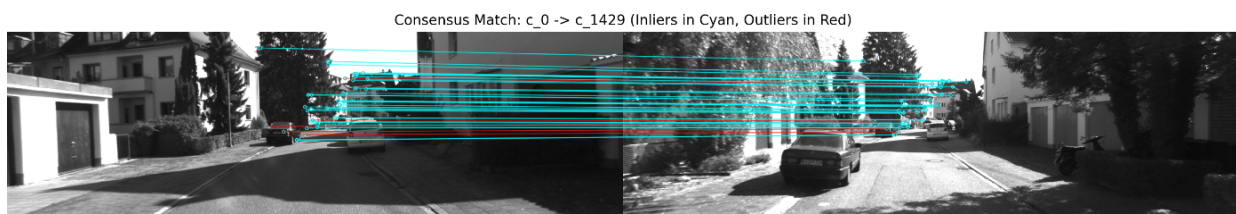


Figure 1: Consensus match between frames 0 and 1429. Inlier matches are shown in cyan and outlier matches in red.

We plotted four versions of the optimized pose graph: before adding loop closures, after adding the first loop-closure factor, after adding approximately half of the selected loop

closures, and after adding all selected loop closures. In our run, these stages correspond to 0, 1, 6, and 10 loop closures, respectively. The resulting trajectories and location covariance ellipses are shown in Figure 2.

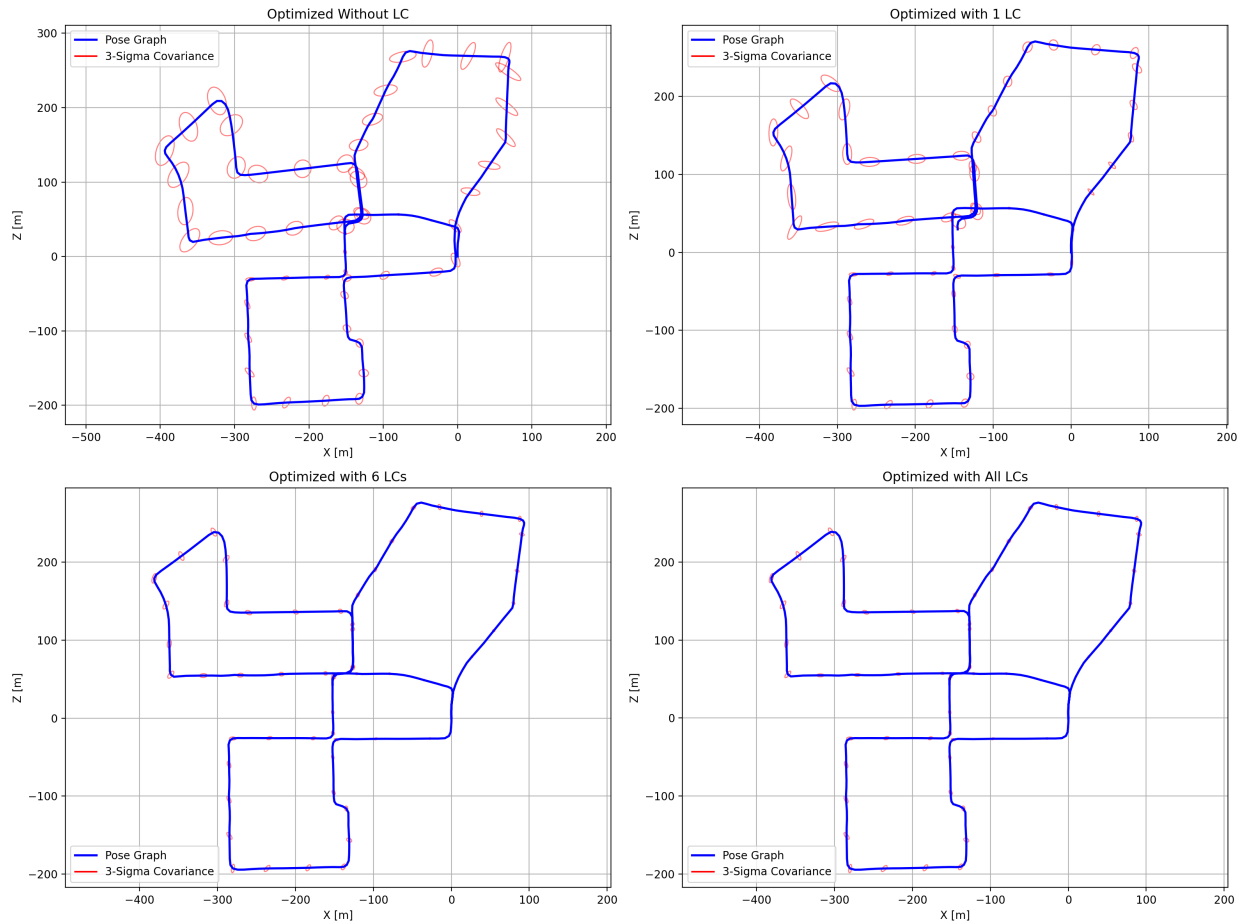


Figure 2: Optimized pose-graph trajectory at four stages of the loop-closure update: before adding loop closures, after 1 loop closure, after 6 loop closures, and after all 10 loop closures. Location covariance ellipses are plotted along the trajectory. The figure was generated by `plot_pose_graphs_versions`, called from `q_5` in `ex7.py`.

We compare the estimated pose-graph trajectory before and after adding loop closures against the ground-truth trajectory in Figure 3. The trajectory with loop closures is visibly closer to the ground truth, showing that the loop-closure constraints significantly reduce the accumulated drift.

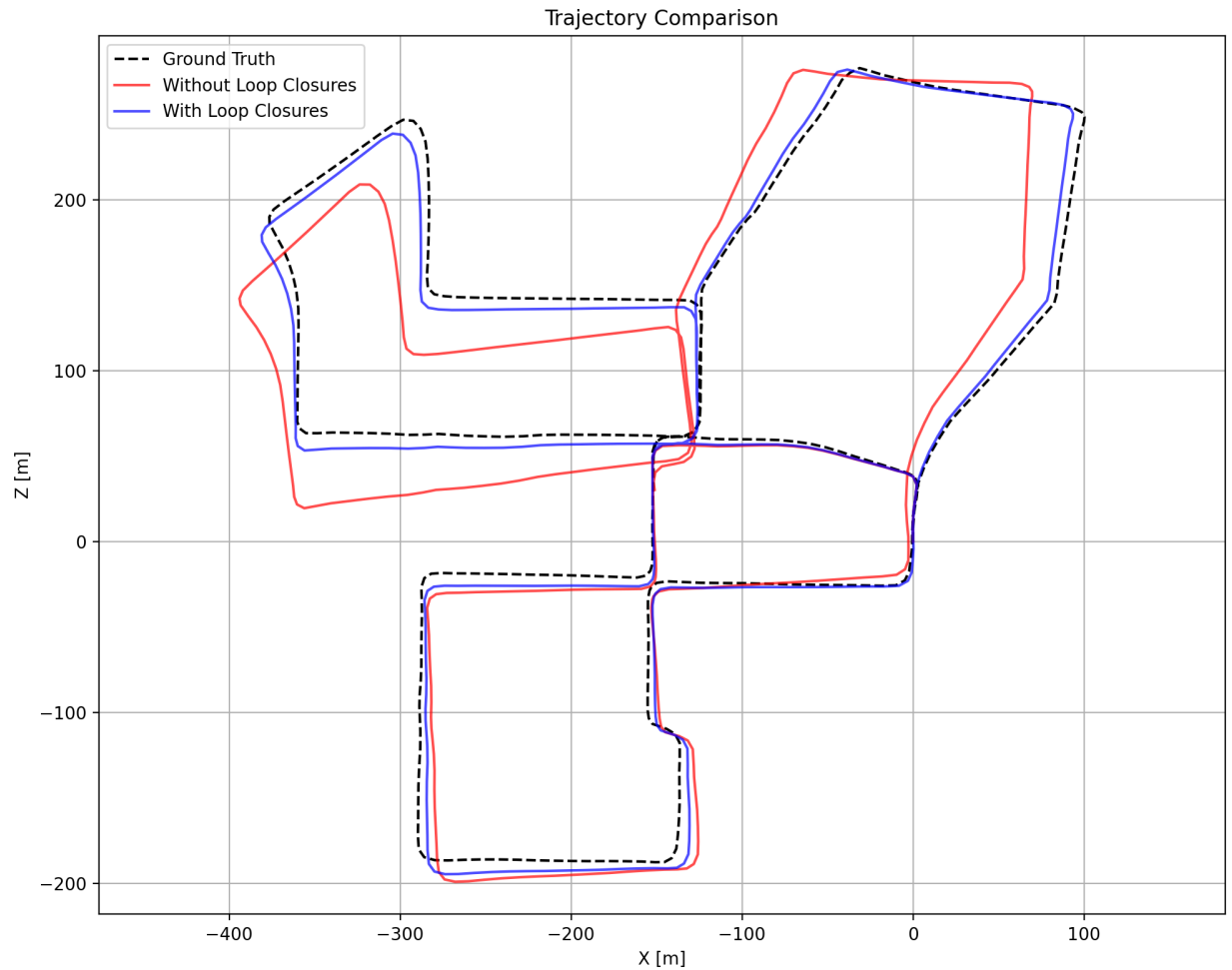


Figure 3: Pose-graph trajectory comparison with and without loop closures, plotted together with the ground-truth trajectory. The figure was generated by `plot_pose_graph_comparisons`, called from `q_5` in `ex7.py`.

We computed the absolute location error of each keyframe by comparing the estimated camera position to the ground-truth position. Figure 4 compares the error before and after adding loop closures. The loop closures reduce the accumulated error: the mean error decreased from 22.28 m to 6.84 m, and the final-frame error decreased from 23.02 m to 4.88 m.

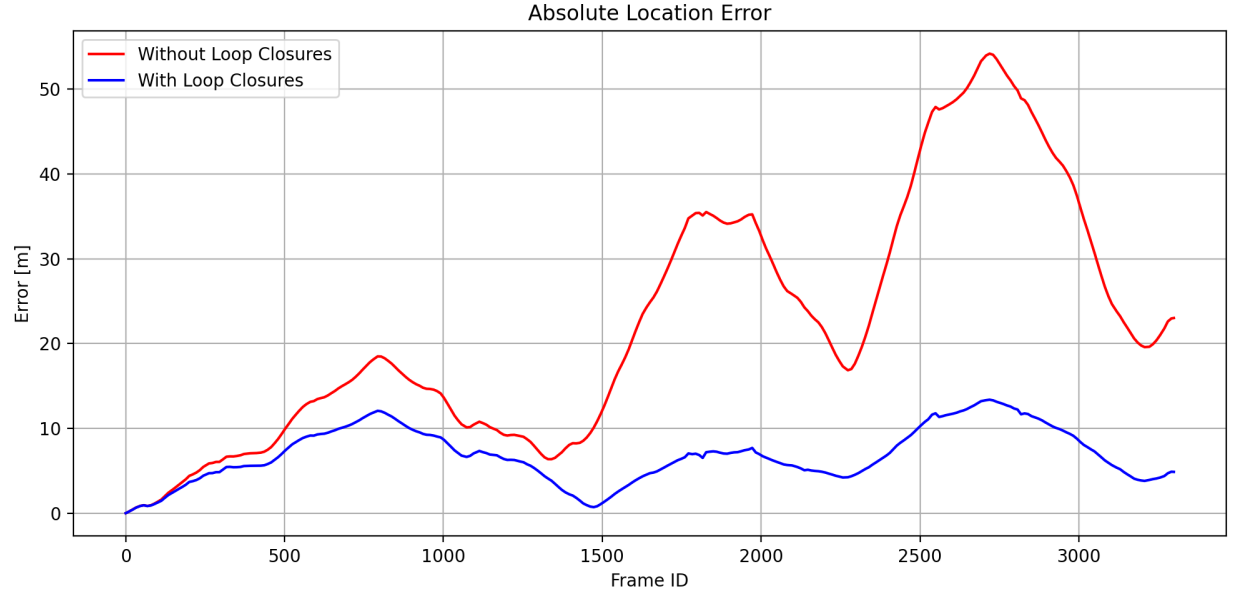


Figure 4: Absolute location error along the pose graph, with and without loop closures. The figure was generated by `plot_absolute_location_error`, called from `q_5` in `ex7.py`.

We measured the location uncertainty size using the determinant of the marginal translation covariance matrix, $\det(\Sigma_t)$, where Σ_t is the 3×3 translation block of the marginal pose covariance. This measure represents the volume of the positional uncertainty ellipsoid. Figure 5 compares this uncertainty measure before and after adding loop closures.

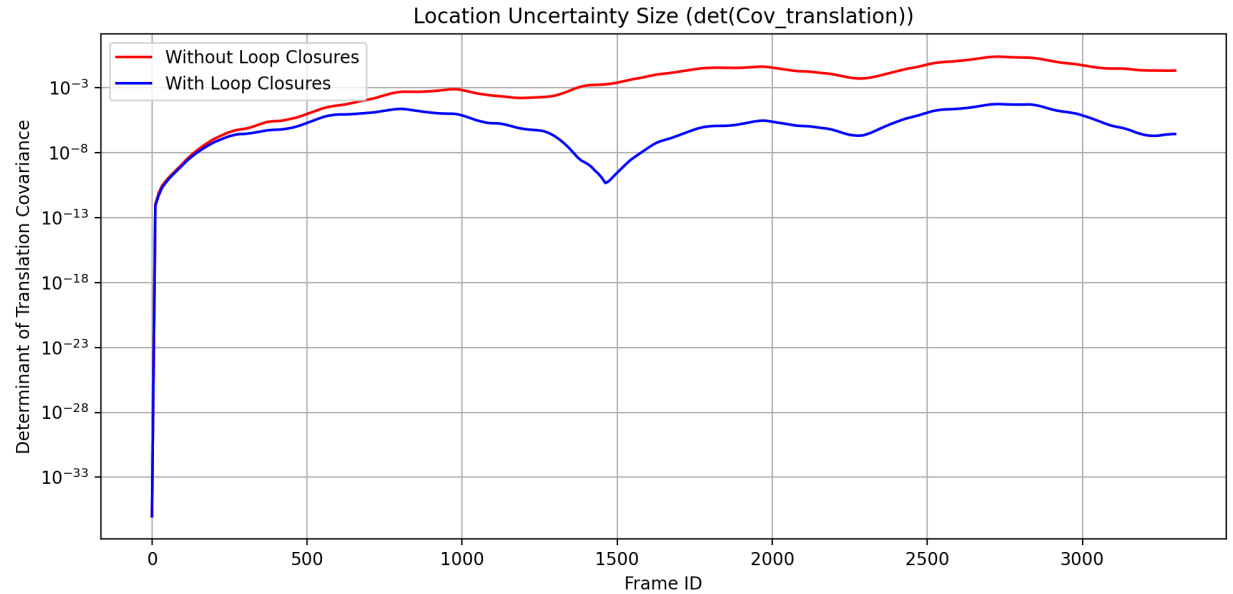


Figure 5: Location uncertainty size along the pose graph, with and without loop closures. The uncertainty measure is $\det(\Sigma_t)$, where Σ_t is the marginal translation covariance. The figure was generated by `plot_location_uncertainty`, called from `q_5` in `ex7.py`.